SERIES — S

12V Sodium-Ion Battery

12V	30–100Ah	3K+	95%	−40/+60°C	Zero	10yr+	IP65
NOMINAL VOLTAGE	CAPACITY RANGE	CYCLE LIFE	ROUND-TRIP EFF.	OPERATING TEMP	MAINTENANCE	DESIGN LIFE	ENCLOSURE RATING

ENERGY STORAGE YOU CAN BANK ON

www.CoulombTechnology.com · sales@coulombtechnology.com · 208-768-8888

PRODUCT OVERVIEW

Series S 12V Battery

- Sodium-Ion NFPP
- Drop-In Lead-Acid Replacement
- 30 / 60 / 100 Ah Models

The Series S 12V is a direct drop-in replacement for lead-acid batteries across mobility, off-grid, marine, and commercial applications. Built on Coulomb Technology's Sodium-Ion NFPP chemistry, it delivers **10× longer cycle life**, zero maintenance, and a 40% weight reduction — with no thermal runaway risk and no toxic materials.

Available in three capacity configurations:

- Series S 12V — 30Ah (360 Wh)
- Series S 12V — 60Ah (720 Wh)
- Series S 12V — 100Ah (1.2 kWh)
- Integrated BMS with cell balancing
- Standard BCI group form factors
- Bluetooth monitoring (optional)

KEY BENEFITS

Zero Thermal Runaway

Sodium-Ion NFPP chemistry is inherently non-flammable — no thermal runaway risk, no toxic gas emissions, and no special shipping restrictions.

- ✓ Air-shippable — no hazmat restrictions
- ✓ No hydrogen gas emissions
- ✓ Non-toxic, fully recyclable
- ✓ Simpler installation approvals

Extreme Temperature Range

Operates from −40°C to +60°C without derating — outperforming lead-acid in cold climates and high-heat environments alike.

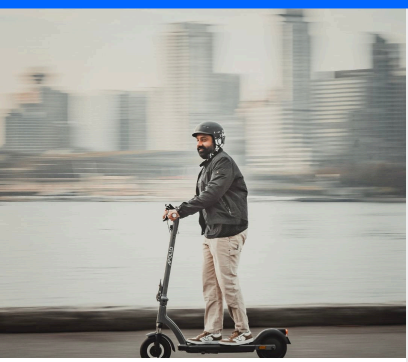
- ✓ −40°C to +60°C discharge range
- ✓ No capacity loss in cold weather
- ✓ 3,000+ cycles at 80% DoD
- ✓ 10+ year design life

True Drop-In Replacement

Standard 12V form factor with BCI-compatible dimensions — replaces lead-acid with no system modifications, no new chargers, no rewiring.

- ✓ BCI group-compatible dimensions
- ✓ Works with existing chargers
- ✓ Zero maintenance — no watering
- ✓ 40% lighter than lead-acid

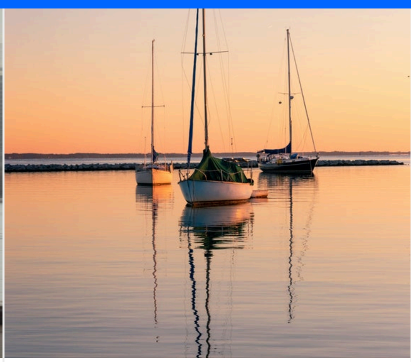
WHERE THE SERIES S PERFORMS



GOLF CARTS / SCOOTERS / LIGHT EV

Mobility & Light EV

Direct replacement for lead-acid in golf carts, electric scooters, and light electric vehicles — with 3× the range per charge and zero maintenance.



MARINE / RV / OFF-GRID

Marine & Off-Grid Power

Lightweight, vibration-resistant, and fully sealed for marine, RV, and off-grid solar applications — with no outgassing and no watering required.

FORKLIFTS / WAREHOUSE / LOGISTICS

Material Handling

Fast charging and extreme temperature tolerance for 24/7 warehouse operations — forklift fleets, pallet jacks, and automated guided vehicles.

TELECOM / UPS / CRITICAL BACKUP

Backup & Telecom Power

Reliable backup power for telecom towers, UPS systems, and critical infrastructure — with 10+ year design life and zero maintenance overhead.

TECHNICAL DATA SHEET

Specifications

coulomb TECHNOLOGY

SERIES S — 12V BATTERY

NOMINAL VOLTAGE	CAPACITY RANGE	CYCLE LIFE
12V	**30–100Ah**	**3K+**
12.8V nominal (4S NFPP)	Three models available	At 80% DoD, 25°C

WEIGHT (100AH)	OPERATING TEMP	EFFICIENCY
13kg	**−40/+60°C**	**95%**
40% lighter than lead-acid	Discharge range	Round-trip efficiency

PARAMETER	VALUE / DESCRIPTION	UNIT
NOMINAL VOLTAGE	12.8 (4S configuration)	VDC
CAPACITY — 30AH MODEL	30 (360 Wh)	Ah
CAPACITY — 60AH MODEL	60 (720 Wh)	Ah
CAPACITY — 100AH MODEL	100 (1,200 Wh)	Ah
USABLE CAPACITY	80–100% DoD (vs. 50% lead-acid)	—
CYCLE LIFE	3,000+ @ 80% DoD, 25°C	—
CELL TYPE	Sodium-Ion NFPP (Non-Flammable Polymer Pouch)	—
DISCHARGE TEMP RANGE	−40°C to +60°C	—
CHARGE TEMP RANGE	−10°C to +45°C	—
SELF-DISCHARGE RATE	<3% per month	—
ROUND-TRIP EFFICIENCY	~95%	—
BMS COMMUNICATION	Integrated BMS; optional Bluetooth monitoring	—
ENCLOSURE RATING	IP65 — sealed, vibration-resistant	—

CERTIFICATIONS

UL 1973 | UL 9540 | UN 38.3

Cell & Module Level

UL 9540A TESTING

Cell: Tested & Passed

Module/System: In Progress

WARRANTY

Cell: 3 Years

System: 5 Years

Extended warranty available

QUALITY ASSURANCE & TEST SCOPE

- ✓ Visual, Mechanical & Dimensional Inspection
- ✓ 3 × 0.2C Capacity Verification (80% DoD)
- ✓ BMS Protections & Isolation Tests
- ✓ Baseline OCV & Self-Discharge Measurement
- ✓ Functional & Conditioning Cycles
- ✓ Vibration & IP65 Seal Verification

TECHNOLOGY COMPARISON

Sodium-Ion NFPP vs. Lead-Acid

- Performance
- Safety
- Total Cost of Ownership

coulomb TECHNOLOGY

WHY SODIUM-ION NFPP OUTPERFORMS LEGACY LEAD-ACID TECHNOLOGY

A Clear Advantage Across Every Performance Dimension

Coulomb Technology's Sodium-Ion NFPP chemistry delivers superior cycle life, wider operating temperature range, zero maintenance, and a fundamentally safer chemistry — making it the ideal replacement for lead-acid in critical commercial and industrial applications.

FEATURE	SODIUM-ION NFPP	LEAD-ACID
Cycle Life	3,000+ cycles @ 80% DoD	300–500 cycles @ 50% DoD
Weight (100Ah)	13 kg (28.7 lb)	~22 kg (48.5 lb)
Usable Capacity	80–100% DoD	50% DoD (to preserve life)
Discharge Temp Range	−40°C to +60°C	−20°C to +50°C
Charge Temp Range	−10°C to +45°C	0°C to +40°C
Maintenance	Zero maintenance	Regular watering & equalization
Self-Discharge	<3% per month	5–15% per month
Charging Efficiency	~95%	~75%
Thermal Runaway Risk	Zero risk	Low risk
Toxic Gas Emissions	None	Hydrogen gas during charging
Memory Effect	None	Sulfation if undercharged
Lifespan	10+ years	3–5 years
Environmental Impact	Non-toxic, recyclable	Lead & acid hazardous

10×
Longer cycle life vs. lead-acid

40%
Lighter weight per 100Ah

Zero
Maintenance required

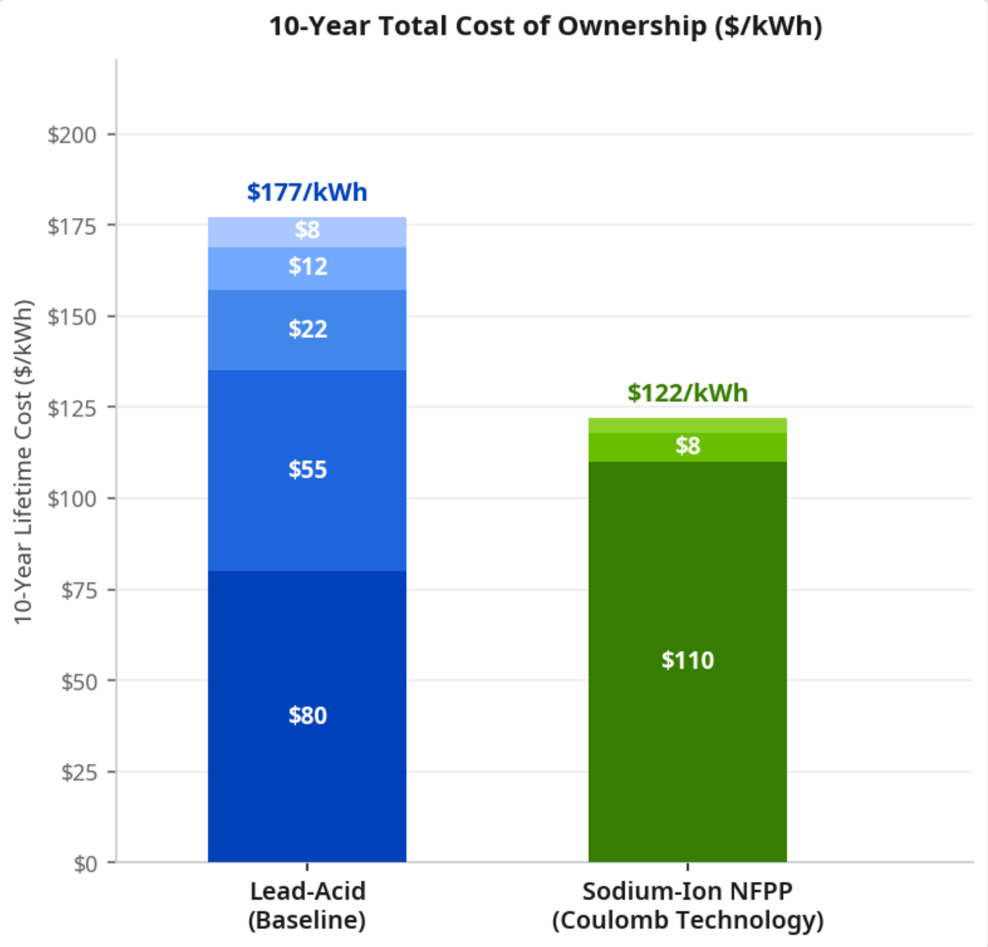
95%
Round-trip efficiency

coulomb TECHNOLOGY

SYSTEM ADVANTAGE OVER LEAD-ACID

31% Lower Total Cost of Ownership Over 10 Years

Sodium-Ion NFPP eliminates replacement costs, maintenance overhead, and hazardous disposal — delivering a 31% lower 10-year TCO vs. lead-acid, validated by real-world deployments.



31% Lower TCO — $122/kWh (NFPP) vs. $177/kWh (Lead-Acid) over 10 years. Savings driven by eliminated replacements, zero maintenance, and lower disposal costs.

31% — Lower 10-year TCO vs. lead-acid ($122 vs. $177/kWh)
Zero — Maintenance cost — no watering, no equalization
10yr+ — Design life vs. 3–5 years for lead-acid

MODEL ASSUMPTIONS

Lead-acid replacement cycle	3–5 yr
NFPP design life	10+ yr
Analysis period	10 yr
Lead-acid replacements	2×
Maintenance cost delta	100%
Weight savings (100Ah)	40%
Charging efficiency delta	20%
Discount rate / NPV calc	8%

COST COMPONENTS

- Lead-Acid — CapEx
- Lead-Acid — Replacement
- Lead-Acid — O&M
- Lead-Acid — Disposal
- NFPP — CapEx
- NFPP — O&M
- NFPP — Disposal
- Lead-Acid — Downtime

RESEARCH & DEVELOPMENT PARTNERS

Oak Ridge National Laboratory · Argonne National Laboratory · Rutgers · Nuvation Energy · Spark Cleantech Accelerator · NENY new energy new york · NEI Corporation · Clean Tech Open 2024 Alumni · Innovation Crossroads · cradle to commerce